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Editorial Office

Journal of Animal Science and Biotechnology

Dear Editors,

We are pleased to submit our manuscript entitled “A Multi-Tissue Transcriptomic Atlas of Porcine Development Identifies Conserved Molecular Programs with Humans” for consideration as a Research Article in the *Journal of Animal Science and Biotechnology*.

A fundamental challenge in pig research is the lack of a standardised molecular framework for defining developmental stage. Current practice relies on chronological age or morphological criteria, which are poor proxies for biological maturity and introduce confounding variability across studies in nutrition, growth physiology, and disease modelling. This gap limits reproducibility and hinders cross-species translation—despite the pig being one of the most important preclinical models for human conditions.

To address this, we constructed the first multi-tissue transcriptomic atlas of porcine postnatal development. Using 1,924 RNA-seq profiles from the PigGTEx resource spanning five major tissues (brain, liver, muscle, lung, and blood), we trained ordinal machine learning models that infer developmental stage directly from gene expression with high accuracy (mean balanced accuracy: 0.90). Our analysis reveals profoundly tissue-specific molecular programs underlying maturation, with extremely low cross-tissue feature overlap (0.2% Jaccard similarity), demonstrating that each tissue follows distinct regulatory logic during development.

To assess the biological relevance of our framework, we performed a preliminary cross-species comparison with human skeletal muscle, identifying 66 conserved developmental genes with 97% directional concordance in expression trajectories (Pearson $r = 0.69$, 95% CI: 0.54–0.80). These genes organise into three functional modules—neuromuscular maturation, structural refinement, and metabolic transition—comprising well-characterised developmental regulators such as *SORCS2*, *FGFRL1*, and *IGF2BP2/3*.

We believe this work is well suited for the *Journal of Animal Science and Biotechnology* for three reasons. First, the atlas provides an immediately applicable tool for standardising experimental cohorts in swine nutrition, production, and biomedical research—topics central to the journal's scope. Second, by demonstrating conserved molecular programs between pigs and humans, we strengthen the scientific basis for using pigs as translational models, with direct implications for disease modelling (e.g., Duchenne Muscular Dystrophy). Third, the framework is designed to be species-agnostic and extensible to other livestock species as developmental transcriptomic data become available, aligning with the journal's broad interest in animal biotechnology.

This manuscript has not been published previously and is not under consideration elsewhere. All authors have approved the submitted version. We have no conflicts of interest to declare.

We suggest the following potential reviewers based on their expertise in porcine genomics, developmental biology, and cross-species translation:

1. Prof. Emeritus Martien Groenen — Wageningen University & Research, Netherlands (pig genome assembly, comparative genomics, FAANG steering committee)
2. Prof. Christopher Tuggle — Iowa State University, USA (porcine functional genomics, immunogenomics, FAANG consortium)
3. Dr Kyle Schachtschneider — University of Illinois at Chicago / Sus Clinicals Inc., USA (porcine epigenetic clocks, biological aging, pig as biomedical model)

Thank you for considering our submission. We look forward to your response.

Yours sincerely,

Peter Theobald

on behalf of all authors

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